FLAIR Task Schematic

Unplanned ICU Readmission (Task T5)

FLAIR Benchmark

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Abstract

We specify the FLAIR unplanned ICU readmission task: at the end of a patient's first ICU stay, predict whether the patient will return to the ICU later in the *same* hospitalization through an unplanned path. The signal is derived entirely from the CLIF `ADT` location history. A *planned* return — ICU → Procedural → ICU, e.g. a trip to the OR or cath lab and back — is not counted: the procedural segment is removed and the two ICU stays merge into one. The task is one-per-stay (exactly one prediction per surviving hospitalization), in contrast to the continuous, one-per-timestep FLAIR tasks.

1 FLAIR operational definition

1.1 Anchors and windows

Let H_i denote a hospitalization. From the CLIF `ADT` table we compute (see `compute_readmission_timing`) a merged location sequence and, from it, the first ICU stay window $[t_i^{\mathrm{ICU}_1,\mathrm{start}}, t_i^{\mathrm{ICU}_1,\mathrm{end}}]$ and the start time of a second ICU stay (if any).

Location standardization.
Each `adt.location_category` is lowercased and mapped to a coarse bucket `std_location` $\in$ {ICU, Ward, ER, Procedural, Other} via `LOCATION_MAPPING`. Stepdown, L&D, hospice, psych, rehab, radiology, dialysis, and other all map to `Other`.

Procedural-sandwich removal (planned readmission).
A row with `std_location` = Procedural whose immediate predecessor and successor (within H_i, ordered by `in_dttm`) are both ICU is deleted. This collapses ICU → Procedural → ICU into a single ICU stay: a planned procedural excursion is *not* a readmission.

Consecutive-location merge.
After sandwich removal, consecutive rows with the same `std_location` are merged into one location group spanning [min(`in_dttm`), max(`out_dttm`)]. Back-to-back ICU ADT rows therefore form a single ICU stay.

First / second ICU stay.
Rank the merged ICU groups of H_i by `in_dttm`. Group 1 yields $t_i^{\mathrm{ICU}_1,\mathrm{start}}, t_i^{\mathrm{ICU}_1,\mathrm{end}}$. Group 2, if present, is the unplanned readmission with start `second_icu_start_time`.

Prediction time.
prediction_dttm$_i = t_i^{\mathrm{ICU}_1,\mathrm{end}}$ (decision made at first-ICU discharge).

Input window.
$[\,\text{window_start}_i, \text{window_end}_i\,] = [t_i^{\mathrm{ICU}_1,\mathrm{start}}, t_i^{\mathrm{ICU}_1,\mathrm{end}}]$ — the entire first ICU stay (variable length; no fixed lookback).

Label (`label_icu_readmission`).
$y_i = \mathbb{1}[\texttt{second_icu_start_time}_i \neq \varnothing]$.

1.2 Cohort

- Age at admission ≥ 18 with valid `admission_dttm` and `discharge_dttm`.
- ≥ 1 ICU ADT record (first ICU stay defined).
- **Exclude** hospitalizations with no ICU data (no first ICU stay) — realized as an inner join with the timing table.
- **Exclude** death during the first ICU stay (`death_dttm` $\in [t_i^{\mathrm{ICU}_1,\mathrm{start}}, t_i^{\mathrm{ICU}_1,\mathrm{end}}]$): the readmission outcome is undefined.
- **Exclude** direct ICU discharge (the last merged location is ICU): there is no opportunity to readmit.
- **Exclude** a first ICU stay shorter than `MIN_ICU_STAY_HOURS` = 1 hour.
- **Exclude** an intermediate ER visit (an ER location after the first location), which confounds the planned/unplanned distinction.

1.3 CLIF tables and categories used

Task T5 reads three CLIF v2.1 tables. The entire readmission signal is the `adt.location_category` sequence; `hospitalization` and `patient` provide cohort filters, the death timestamp, and demographics.

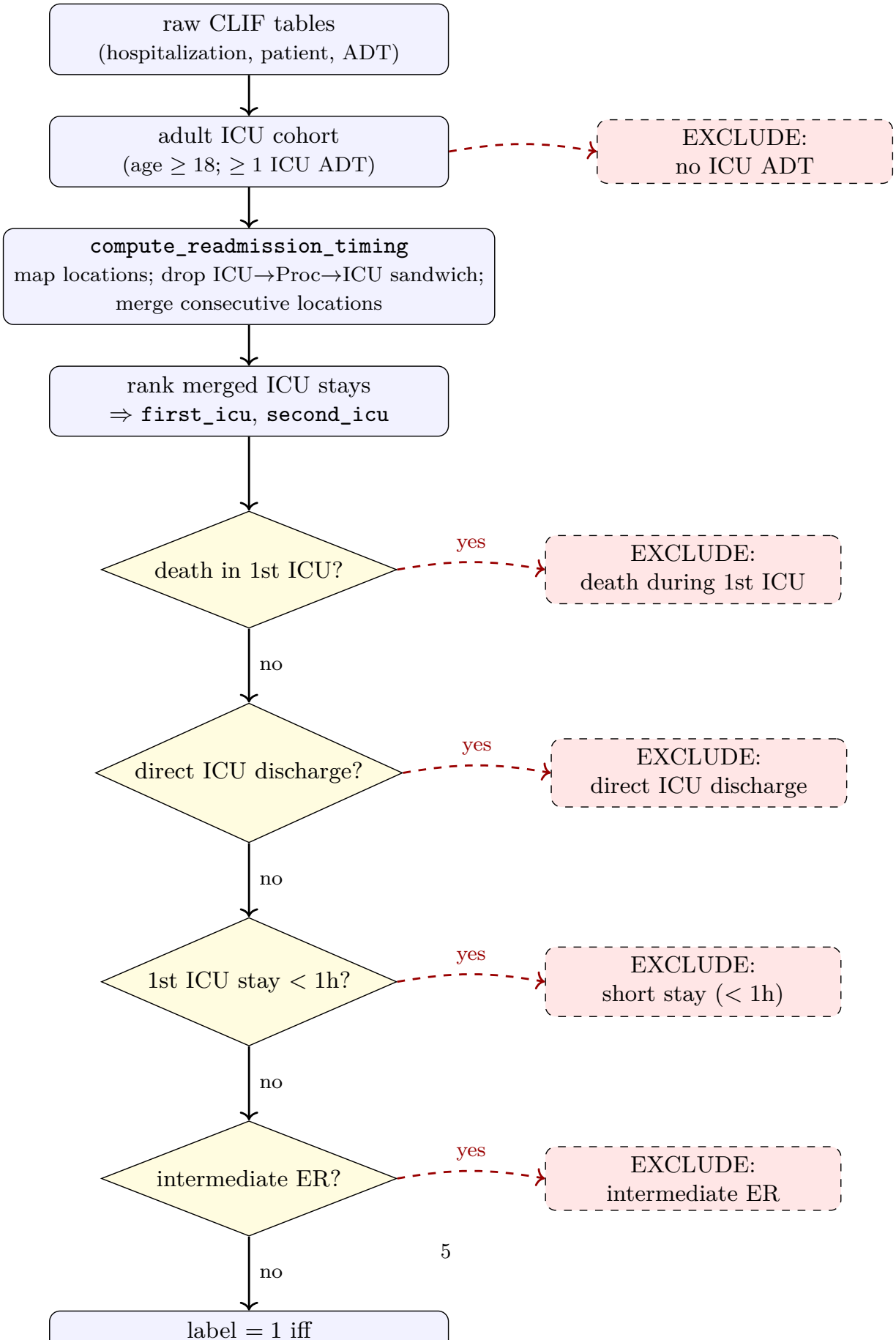
CLIF table	Category column	Use
`hospitalization`	—	cohort/demographics join; `age_at_admission`, valid dates
`patient`	—	demographics join; `death_dttm` for the death exclusion
`adt`	`location_category`	merged into ICU / Ward / ER / Procedural / Other; sole readmission signal

Table 1: CLIF surface for Task T5. No `labs`, `vitals`, or medication tables are read — the task is ADT-only.

Columns selected from each table:

- hospitalization: hospitalization_id, patient_id, admission_dttm, discharge_dttm, age_at_admission, discharge_category.
- patient: patient_id, sex_category, race_category, ethnicity_category, death_dttm.
- adt: hospitalization_id, location_category, in_dttm, out_dttm.

2 Algorithmic flow



Step-by-step prose:

1. Load CLIF tables: `hospitalization`, `patient`, `adt` via `base_icu_cohort`.
2. Apply adult-ICU cohort filters (age, valid admission/discharge, ≥ 1 ICU ADT).
3. `compute_readmission_timing(adt)`: map locations; drop ICU $\rightarrow$ Procedural $\rightarrow$ ICU procedural rows; merge consecutive same-location rows; rank ICU groups to extract first/second ICU times; flag `is_direct_icu_discharge` and `has_intermediate_er`.
4. Inner-join timing into the cohort (drops no-ICU hospitalizations).
5. Apply the four remaining exclusions: death during first ICU stay; direct ICU discharge; first ICU stay $<$ 1h; intermediate ER.
6. Label $y_i = \mathbb{1}[\texttt{second_icu_start_time} \neq \varnothing]$.
7. Set `prediction_dttm` $= t_i^{\mathrm{ICU}_1,\mathrm{end}}$ and the input window $[t_i^{\mathrm{ICU}_1,\mathrm{start}}, t_i^{\mathrm{ICU}_1,\mathrm{end}}]$.
8. Assign temporal train/test split via `flair_benchmark._split.assign_split`, keyed on `prediction_dttm`.
9. Select `OUTPUT_COLS` (the standard 10-column schema) and persist as parquet.

3 Design choices and rationale

Why ADT-only. ICU readmission is a movement event, fully observable from the CLIF `ADT` location stream without labs, vitals, or notes. Restricting the label to ADT keeps it portable across every CLIF site and avoids the cadence/vocabulary assumptions that complicate the lab- and medication-anchored tasks. The federated ML literature treats unplanned ICU readmission as a structured-EHR target in exactly this spirit (Rojas et al., 2018).

Why remove the procedural sandwich. A patient who leaves the ICU for a scheduled procedure (OR, cath lab, interventional radiology) and returns has not been *readmitted* in any clinically meaningful sense — the return was planned at the time of transfer. Counting it would inflate the positive rate with non-events and penalize a model for "missing" an outcome that carries no deterioration signal. Collapsing ICU $\rightarrow$ Procedural $\rightarrow$ ICU to a single stay encodes the planned/unplanned distinction at the cohort level. Procedural rows that are *not* bracketed by ICU on both sides are left in place.

Why one-per-stay (not continuous). Unlike the deterioration tasks, readmission has a single natural decision point — the moment of ICU discharge, when a clinician decides the patient is stable enough to leave. A one-per-stay framing matches that decision and keeps the label balanced (each hospitalization contributes one row rather than many trivially-negative hourly rows).

Why the four post-cohort exclusions. Death in the first ICU stay makes the outcome undefined; a direct ICU discharge removes any opportunity to readmit (a structurally guaranteed negative); a sub-hour first ICU stay is typically an administrative artifact rather than a true ICU course; an intermediate ER visit muddies the planned/unplanned reading of a subsequent ICU entry. Each is a deterministic ADT/timestamp predicate, applied in a fixed order.

4 Out of scope

- Cross-hospitalization (30-day) ICU or hospital readmission.
- Third and later ICU readmissions within a stay.
- Time-to-readmission (survival) labels.
- ICD- or problem-list–based planned/unplanned adjudication (we use ADT location categories only).
- Input feature design.

References

Juan C. Rojas, Kyle A. Carey, Dana P. Edelson, Laura R. Venable, Michael D. Howell, and Matthew M. Churpek. Predicting intensive care unit readmission with machine learning using electronic health record data. *Annals of the American Thoracic Society*, 15(7):846–853, 2018. doi: 10.1513/AnnalsATS.201710-787OC.